

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Conservation priorities, legislation & protocols - Set 1

### Mark scheme

AQA spec 3.1.2.3.1-3.1.2.3.2; The living environment; textbook pp. 26-39

### Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>01.1</b> | <p><b>1 mark each</b><br/>1 mark for each of the 3 points below.</p> <p><b>1</b> Coordinating global data on biodiversity conservation (Red List)<br/> <b>1</b> Increasing understanding of the importance of biodiversity<br/> <b>1</b> Deploying nature-based solutions to global challenges in climate, food and sustainable development</p> <p><b>Also accept</b></p> <ul style="list-style-type: none"> <li>• Any other correct, relevant point of the same kind - wording never has to match these notes.</li> <li>• <b>biodiversity</b> → the variety of life: habitat, species and genetic diversity</li> <li>• <b>sustainable</b> → meeting present needs without stopping future generations meeting theirs</li> </ul> <p><b>Do not credit</b></p> <ul style="list-style-type: none"> <li>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.</li> <li>• 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future</li> </ul> | <b>3</b> |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks |
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| <b>01.2</b> | <p><b>1 mark each</b><br/>1 mark for each of the 3 points below.</p> <p><b>1</b> Decline = <math>(12\,000 - 2100) / 12\,000 \times 100 = 82.5\%</math></p> <p><b>1</b> Decline over 80% meets the Critically Endangered criterion even though habitat area (<math>400\text{ km}^2</math>) and adults (2100) only meet Endangered</p> <p><b>1</b> A species is placed in the highest category for which it meets any one criterion, so Critically Endangered</p> <p><b>Also accept</b></p> <ul style="list-style-type: none"> <li>Any other correct, relevant point of the same kind - wording never has to match these notes.</li> </ul>                                                                                                                                                                                                                                     | 3     |
| <b>01.3</b> | <p><b>1 mark each</b><br/>1 mark for each of the 3 points below.</p> <p><b>1</b> Categorising a species requires a great deal of research (population, range, trends) and resources<br/>– about 85 000 of millions of species assessed</p> <p><b>1</b> Key species in habitats under particular threat (eg lemurs indicating the state of Madagascar's forests)</p> <p><b>1</b> Evolutionarily unique species / EDGE species<br/>– endemic species<br/>– keystone species<br/>– flagship species<br/>– species with fragmented populations</p> <p><b>Also accept</b></p> <ul style="list-style-type: none"> <li>Any other correct, relevant point of the same kind - wording never has to match these notes.</li> <li><b>endemic</b> → found naturally only in that one place</li> <li><b>keystone species</b> → a species whose loss changes the whole community</li> </ul> | 3     |
| <b>01.4</b> | <p>1 mark for the point below.</p> <p><b>1 Iberian lynx:</b> Critically Endangered to Endangered after captive breeding and protection of two habitat areas (population trebled in 15 years)<br/>– Fregate Island beetle after rat eradication</p> <p><b>Also accept</b></p> <ul style="list-style-type: none"> <li>Any other correct, relevant point of the same kind - wording never has to match these notes.</li> <li><b>captive breeding</b> → breeding in captivity to increase numbers for release</li> </ul> <p><b>Do not credit</b></p> <ul style="list-style-type: none"> <li>captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well</li> </ul>                                                                                                                                                    | 1     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-27; AQA spec 3.1.2.3.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 02.1     | <p><b>2 marks per point</b><br/>2 marks for each of the 4 points below; award 1 mark where a point is only partly made.</p> <p><b>2 EDGE:</b> Evolutionarily Distinct and Globally Endangered - few close relatives, often the last species of their genus / family (Bactrian camel, pygmy hippo, northern bald ibis)<br/>– loss would remove a unique branch of evolution / genes</p> <p><b>2 Endemic:</b> found in only one area (often islands - Galapagos, Madagascar, Seychelles: Aldabra giant tortoise, red ruffed lemur)<br/>– if lost locally there is no population elsewhere to recolonise</p> <p><b>2 Keystone:</b> disproportionately important for the structure of the community relative to its abundance - predation of dominant species, seed dispersal, habitat creation (jaguar, forest elephants keeping paths and water holes open, wolves, beavers)<br/>– losing them causes loss of many other species</p> <p><b>2 Flagship:</b> high public profile (tiger, panda, elephant, orang-utan) raising funds and support that also protects the whole habitat and less charismatic species (fungi, bats, moths)</p> <p><b>Also accept</b></p> <ul style="list-style-type: none"> <li>• Any other correct, relevant point of the same kind - wording never has to match these notes.</li> <li>• <b>Endemic</b> → found naturally only in that one place</li> </ul> | 8     |
| 02.2     | <p><b>1 mark each</b><br/>1 mark for each of the 2 points below.</p> <p><b>1</b> A fragmented population may consist of isolated groups each too small to be viable (resources, inbreeding) even if total numbers are large</p> <p><b>1</b> A species confined to a single area (eg bonobos in one country) is vulnerable to any local change / conflict, unlike chimpanzees spread across many countries</p> <p><b>Also accept</b></p> <ul style="list-style-type: none"> <li>• Any other correct, relevant point of the same kind - wording never has to match these notes.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 28-29; AQA spec 3.1.2.3.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| 03.1     | <p><b>Any 4 of 5</b><br/>1 mark each for any 4 of the 5 points below, in any order, maximum 4.</p> <p><b>1</b> Designation of protected areas (SSSIs, MCZs)</p> <p><b>1</b> Protection of wild birds and their nests (except some pests / game birds)</p> <p><b>1</b> Protection of named mammals such as otters, dormice, red squirrels, badgers and their setts</p> <p><b>1</b> Uprooting wild plants generally illegal</p> <p><b>1</b> Bats must not be disturbed<br/>– roof timber treatments must be non-toxic to bats</p> <p><b>Also accept</b></p> <ul style="list-style-type: none"> <li>• Any other correct, relevant point of the same kind - wording never has to match these notes.</li> </ul> <p><b>Do not credit</b></p> <ul style="list-style-type: none"> <li>• Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
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| 03.2     | <p><b>1 mark each</b><br/>1 mark for each of the 4 points below.</p> <p><b>1 Appendix I:</b> species threatened with extinction - all international (commercial) trade banned, except movement for conservation breeding</p> <p><b>1 Examples:</b> great apes, big cats, rhinos, blue whale</p> <p><b>1 Appendix II:</b> species that may become threatened if trade is not closely controlled - trade permitted under licence from countries where the species is well protected</p> <p><b>1 Examples:</b> Honduras mahogany, hippopotamus, green iguana, great white / basking shark, Venus fly trap</p> <p><b>Also accept</b></p> <ul style="list-style-type: none"> <li>Any other correct, relevant point of the same kind - wording never has to match these notes.</li> <li>The converse of any point, ie the same comparison written the other way round.</li> <li><b>CITES</b> → the treaty that controls international trade in endangered species</li> </ul> <p><b>Do not credit</b></p> <ul style="list-style-type: none"> <li>'CITES bans all trade in wildlife' - it controls and licenses trade, with different rules for each appendix</li> </ul> | 4     |
| 03.3     | <p><b>1 mark each</b><br/>1 mark for each of the 2 points below.</p> <p><b>1</b> Collection may be hard to police in remote areas / poor countries, but if the animals or products cannot be exported / imported there is no market</p> <p><b>1</b> Removing the profit removes the incentive to collect</p> <p><b>Also accept</b></p> <ul style="list-style-type: none"> <li>Any other correct, relevant point of the same kind - wording never has to match these notes.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30-31; AQA spec 3.1.2.3.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks    |
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| <b>04.1</b> | <p><b>1 mark each</b><br/>1 mark for each of the 4 points below.</p> <p><b>1 SSSI:</b> UK legislation (Wildlife and Countryside Act)<br/>         – best examples of UK flora, fauna, geological features<br/>         – management plan agreed with owner who must notify Operations Likely to Damage (ploughing, drainage, pesticides, tree planting / removal)</p> <p><b>1 SPA:</b> EU Birds Directive<br/>         – places of international importance for birds (with SACs under the Habitats Directive form the Natura 2000 network; may restrict activities in surrounding areas)</p> <p><b>1 Ramsar:</b> Ramsar Convention (intergovernmental agreement)<br/>         – internationally important wetlands, originally for waterfowl, now wetland conservation and wise use generally</p> <p><b>1 MCZ:</b> UK marine legislation (Marine and Coastal Access Act) under the W&amp;CA framework<br/>         – protects marine habitats / species by restricting activities such as fishing and dredging<br/>         – NNR: best SSSIs with complete communities</p> <p><b>Also accept</b><br/>         • Any other correct, relevant point of the same kind - wording never has to match these notes.</p> <p><b>Do not credit</b><br/>         • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on<br/>         • quotas described as solving overfishing on their own - undersized fish are still caught and discarded</p> | <b>4</b> |
| <b>04.2</b> | <p><b>1 mark each</b><br/>1 mark for each of the 3 points below.</p> <p><b>1 Legal status:</b> restrictions on damaging activities within the area (and sometimes in surrounding areas, eg drainage)</p> <p><b>1</b> Management agreements between owners and statutory bodies to maintain suitable conditions</p> <p><b>1</b> But it does not protect against external influences such as air pollution or climate change<br/>         – very few sites are state-owned so protection depends on owner cooperation</p> <p><b>Also accept</b><br/>         • Any other correct, relevant point of the same kind - wording never has to match these notes.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>3</b> |
| <b>04.3</b> | <p><b>1 mark each</b><br/>1 mark for each of the 3 points below.</p> <p><b>1</b> Much of the UK's wildlife lives in habitats created by farming (hay meadows, hedgerows) which only survive if the farming continues</p> <p><b>1</b> Managing land for wildlife costs money / reduces income so farmers need payments (points for beetle banks, hedgerows, buffer strips, wild bird seed, low-input grassland)</p> <p><b>1</b> Benefits everyone so society pays through central funding<br/>         – legislation cannot compel positive management</p> <p><b>Also accept</b><br/>         • Any other correct, relevant point of the same kind - wording never has to match these notes.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>3</b> |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30, 38-39; AQA spec 3.1.2.3.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
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| 05.1     | <p><b>Calculation</b><br/>1 mark for the correct answer; working does not have to be shown.</p> <p><b>1</b> <math>(240\,000 - 2\,000) / 240\,000 \times 100 = 99.2\%</math></p> <p><b>Also accept</b></p> <ul style="list-style-type: none"> <li>• The correct answer however it is set out, including a different but valid method.</li> <li>• An answer carried forward from an earlier incorrect step (error carried forward).</li> <li>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.</li> <li>• The value alone - the unit (%) is already printed on the answer line.</li> </ul> <p><b>Do not credit</b></p> <ul style="list-style-type: none"> <li>• A bare wrong answer with no working.</li> </ul>                                                                                                                                         | 1     |
| 05.2     | <p><b>1 mark each</b><br/>1 mark for each of the 3 points below.</p> <p><b>1 Whales are k-selected:</b> slow growing, late maturing, one calf every few years, so populations recover very slowly</p> <p><b>1 Technology</b> (harpoon cannons, steam / factory ships) allowed catches far above the sustainable yield within decades</p> <p><b>1 No regulation / open access to the ocean</b><br/>– whales were valuable (oil, spermaceti) and easily found</p> <p><b>Also accept</b></p> <ul style="list-style-type: none"> <li>• Any other correct, relevant point of the same kind - wording never has to match these notes.</li> <li>• <b>sustainable</b> → meeting present needs without stopping future generations meeting theirs</li> </ul> <p><b>Do not credit</b></p> <ul style="list-style-type: none"> <li>• 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future</li> </ul> | 3     |
| 05.3     | <p><b>1 mark each</b><br/>1 mark for each of the 4 points below.</p> <p><b>1 Moratorium on commercial whaling</b> since 1986 until populations recover</p> <p><b>1 Total protection for certain species</b><br/>– whale sanctuaries</p> <p><b>1 Limits on numbers and sizes taken</b><br/>– protection of suckling mothers and calves</p> <p><b>1 Research into whale biology and threats</b> (ship strikes, net entanglement)<br/>– exceptions: aboriginal subsistence quotas (eg Inuit bowheads), 'scientific' permits (Japan - ruled not research in 2014), Norway / Iceland set their own quotas</p> <p><b>Also accept</b></p> <ul style="list-style-type: none"> <li>• Any other correct, relevant point of the same kind - wording never has to match these notes.</li> </ul>                                                                                                                                                                        | 4     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 05.4     | <p><b>1 mark each</b><br/>1 mark for each of the 2 points below.</p> <p><b>1</b> ITTO's 1990 target of trade only from sustainably managed forests by 2000 was not met and little progress since</p> <p><b>1</b> Timber comes from many countries / private land with weak enforcement, high demand and alternative land uses<br/>– whaling involves few nations and a single treaty body</p> <p><b>Also accept</b></p> <ul style="list-style-type: none"> <li>Any other correct, relevant point of the same kind - wording never has to match these notes.</li> <li><b>sustainable</b> → meeting present needs without stopping future generations meeting theirs</li> </ul> <p><b>Do not credit</b></p> <ul style="list-style-type: none"> <li>'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future</li> </ul> | 2     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 31-32; AQA spec 3.1.2.3.2.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |